

A: SEQUENTIAL FILE ALLOCATION

C PROGRAM:

```
#include <stdio.h>

int main()
{
    int start, length, i;

    printf("Enter Starting Block: ");
    scanf("%d", &start);

    printf("Enter File Length (Number of Blocks): ");
    scanf("%d", &length);

    printf("\nAllocated Blocks:\n");
    for(i = 0; i < length; i++)
    {
        printf("%d ", start + i);
    }

    printf("\n");

    return 0;
}
```

SHELL SCRIPT:

```
#!/bin/bash

echo "Enter Starting Block:"
read start

echo "Enter File Length:"
read length

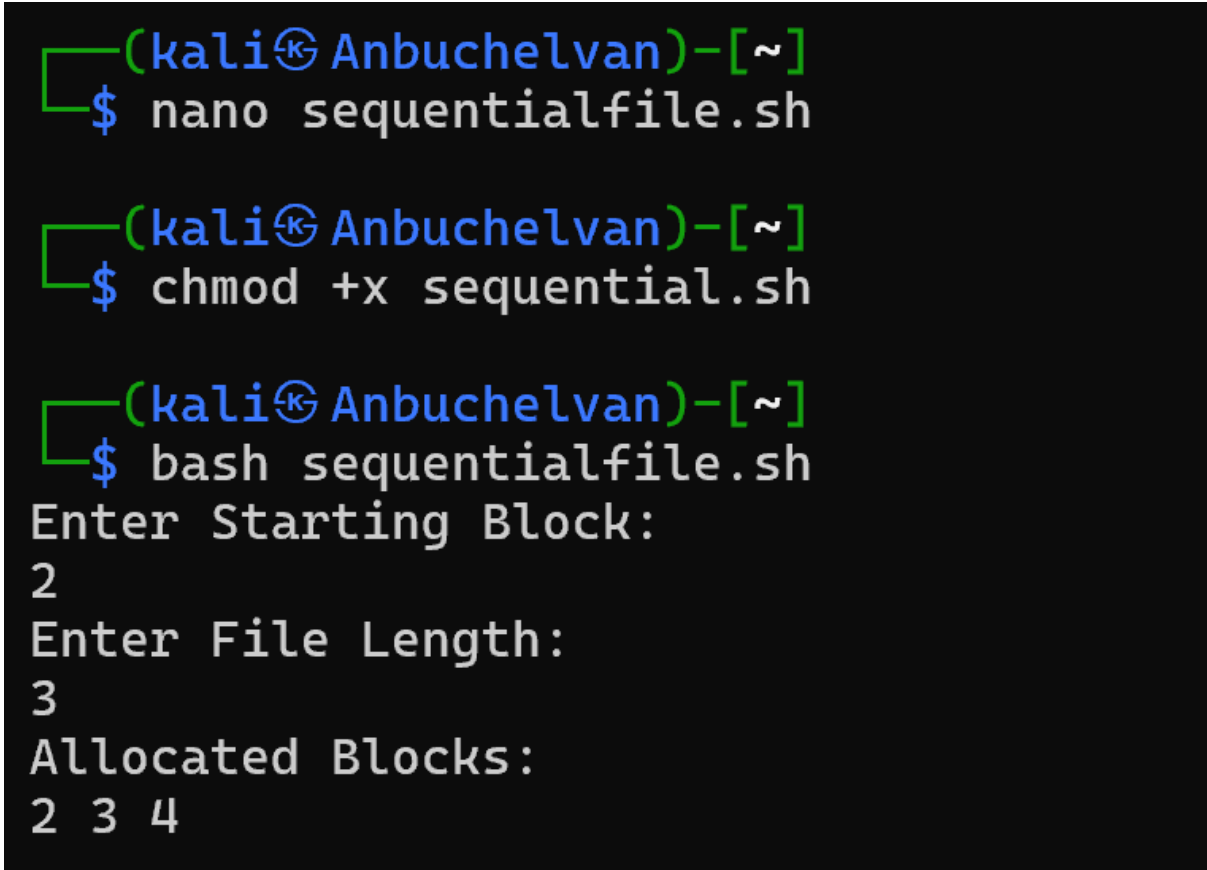
echo "Allocated Blocks:"
for ((i=0;i<length;i++))
do
```

```
echo -n "$((start+i)) "
```

```
done
```

```
echo
```

OUTPUT:



```
(kaliⓀ Anbuchelvan)-[~]  
$ nano sequentialfile.sh  
  
(kaliⓀ Anbuchelvan)-[~]  
$ chmod +x sequential.sh  
  
(kaliⓀ Anbuchelvan)-[~]  
$ bash sequentialfile.sh  
Enter Starting Block:  
2  
Enter File Length:  
3  
Allocated Blocks:  
2 3 4
```

B: INDEXED FILE ALLOCATION

C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n, indexBlock, blocks[20], i;
```

```
printf("Enter Index Block: ");
```

```
scanf("%d", &indexBlock);
```

```
printf("Enter Number of Blocks: ");
```

```
scanf("%d", &n);
```

```

printf("Enter Block Numbers:\n");

for(i = 0; i < n; i++)
{
scanf("%d", &blocks[i]);
}

printf("\nIndex Block : %d\n", indexBlock);

printf("Allocated Blocks : ");

for(i = 0; i < n; i++)
{
printf("%d ", blocks[i]);
}

printf("\n");

return 0;
}

```

SHELL SCRIPT:

```

#!/bin/bash

echo "Enter Index Block:"

read index

echo "Enter Number of Blocks:"

read n

echo "Enter Block Numbers:"

for ((i=0;i<n;i++))
do
read block[$i]
done

echo "Index Block : $index"

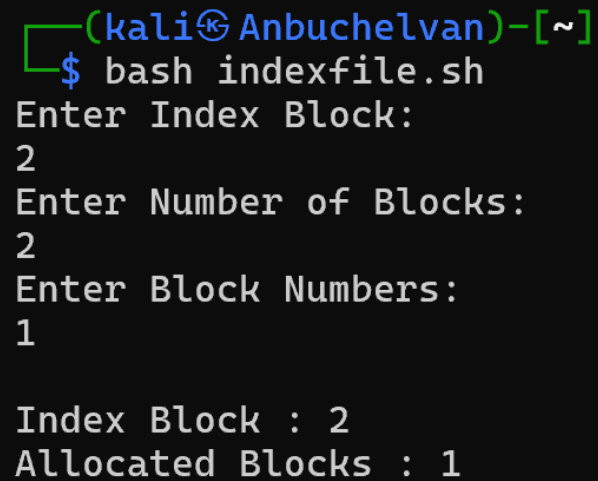
echo -n "Allocated Blocks : "

for ((i=0;i<n;i++))

```

```
do
echo -n "${block[$i]} "
done
echo
```

OUTPUT:

A terminal window with a black background and green text. The prompt is (kali@Anbuchelvan)~. The user enters 'bash indexfile.sh'. The program prompts for 'Enter Index Block:' and the user enters '2'. It then prompts for 'Enter Number of Blocks:' and the user enters '2'. Finally, it prompts for 'Enter Block Numbers:' and the user enters '1'. The program then outputs 'Index Block : 2' and 'Allocated Blocks : 1'.

```
(kali@Anbuchelvan)~  
$ bash indexfile.sh  
Enter Index Block:  
2  
Enter Number of Blocks:  
2  
Enter Block Numbers:  
1  
  
Index Block : 2  
Allocated Blocks : 1
```

C: LINKED FILE ALLOCATION

C PROGRAM:

```
#include <stdio.h>

int main()
{
    int n, blocks[20], i;
    printf("Enter Number of Blocks: ");
    scanf("%d", &n);
    printf("Enter Block Numbers:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &blocks[i]);
    }
    printf("\nLinked Allocation:\n");
```

```
for(i = 0; i < n - 1; i++)
{
printf("%d --> ", blocks[i]);
}
printf("%d --> NULL\n", blocks[n - 1]);
return 0;
}
```

SHELL SCRIPT:

```
#!/bin/bash

echo "Enter Number of Blocks:"

read n

echo "Enter Block Numbers:"

for ((i=0;i<n;i++))
do
read block[$i]
done

echo "Linked Allocation:"

for ((i=0;i<n-1;i++))
do
echo -n "${block[$i]} --> "
done

echo "${block[$((n-1))]} --> NULL"
```

OUTPUT:

```
(kali㉿ Anbuchelvan)-[~]  
$ nano linked.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ chmod +x linked.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ bash linked.sh
```

Enter Number of Blocks:

2

Enter Block Numbers:

3

Linked Allocation:

3 --> --> NULL